NIHR | National Institute for Health and Care Research

Summary

Reference Number	NIHR304711
Lead Applicant	Dr Chris Carvalho
Research Title	Real-World Health and Financial Impact of Medication Optimisation on Kidney and Cardiovascular disease: A Comprehensive Analysis in an Ethnically Diverse UK Urban Population
Total Research Cost	936,127.00

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1. Doctoral Fellowship PPI Assessment Form

1. Doctoral Fellowship PPI Assessment Form

Documents

I confirm that I have read and agree to abide by the Code of Practice
☒Confirmed

A Plain English Summary is a brief summary of a research project or research proposal that has been written for members of the public, rather than for researchers or professionals. It should be written in plain English, avoiding the use of jargon and explaining any technical terms that have to be included. The summary should explain why the research is being suggested, what the researchers aim to achieve, and how this may impact on the field(s) of interest.

If the project is funded, the Plain English Summary will be published as stand-alone descriptive text on a variety of websites, and, as such, needs to serve as a description of the work in full.

Justification of grading provided above Please use the space provided to explain your grading, highlighting strengths and weaknesses of the summary as appropriate.
This is a well written and well structured PES. It was easy to understand in a single reading, although one word is not common (maximally - perhaps 'best' would be simple to substitute). The section headings and text are clear and comprehensive and the style is interesting and informative. A good PES.

All mandatory questions have been completed. If your responses are complete the form can be submitted.

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If the project is funded, the Plain English Summary will be published as stand-alone descriptive text on a variety of websites, and, as such, needs to serve as a description of the work in full.

Justification of grading provided above Please use the space provided to explain your grading, highlighting strengths and weaknesses of the summary as appropriate.
The Plain English Summary is well presented. It is clear and helpful. There is no Jargon. There are no long sentences. There are Headings and Sections which help. The PES could be used on its own to describe the research.

All mandatory questions have been completed. If your responses are complete the form can be submitted.

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1. Review

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Selection Committee Assessment

Please read the application and provide a score for each assessment section using the following scale:

Person
Applicants for a Doctoral Fellowship should meet the following criteria:

- Relevant previous research experience and/or training suitable for undertaking a PhD in applied health or social care research
- Outputs from previous research experience and training relative to career stage and background
- Potential to become a future leader in applied health or social care research

Score (1-6)	5

Reason for your Score Please provide the reason for your score and how it relates to the applicant's performance against the assessment criteria. This information will be sent to applicants as feedback.
Candidate with significant experience and track record in the proposed area of research in this fellowship application. Several publications, but no awards nor grants.

Research Proposal
Applicants for a Doctoral Fellowship should meet the following criteria:

- Scope of research suitable for a PhD
- Suitability and standard of methodology and research design to answer research question being proposed
- Likely impact of the research to patients/service users, carers, the public and health and/or care services
- Quality of doctoral research already undertaken (if applicable and/or where relevant)
- Appropriate and meaningful patient/service users, carer and public involvement
- Quality and scope of review of existing evidence

Score (1-6)	3

Reason for your Score Please provide the reason for your score and how it relates to the applicant's performance against the assessment criteria. This information will be sent to applicants as feedback.
A version of the PhD proposal has been submitted to another scheme, the outcome of which is was expected on 20th February 2024. Do we know what the outcome was? Is Work Package 1, an evidence synthesis of existing systematic reviews, needed? In WP2, a cross-sectional study will assess the current level of the key CKD medication usage in the relevant CKD population and statistically analyse differences by subgroup to assess health inequalities in medication usage. I am not clear why a static picture, such as one provided by a cross sectional study, is the appropriate way to look at this. A longitudinal study, may be more appropriate. Do such data exist? Is the choice of a cross-sectional data set dictated by the only available data on this? WP3, intends to develop a decision model to assess the impact of alternative prescription policies for medicine optimisation. I am not clear where the model input parameters be estimated from and using what methods. The will not come from the analysis of the data in WP2, which seems to be of descriptive nature. The type of what-if analyses that are needed for the decision model and simulation, require longitudinal data.

So, while the research is relevant and of high priority I am not convinced the data available can answer the questions set out in the application.

Good PPI section.

Site and Support
Applicants for a Doctoral Fellowship should meet the following criteria:

- Suitability and experience of supervisory team
- Standard and level of host organisation's support including evidence of supporting other early career researchers
- Track record of Host Department(s) in the research area being proposed
- Evidence of the host organisation(s) commitment to creating and maintaining an inclusive and supportive research culture, including evidence of commitment to the principles of equality, diversity and inclusion and research integrity

Score (1-6)	5

Reason for your Score

Please provide the reason for your score and how it relates to the applicant's performance against the assessment criteria. This information will be sent to applicants as feedback.

Strong supervisory team; they are however quite senior and pretty busy. Need reassurance that they will dedicate sufficient time to the PhD supervision.

Research site with an international reputation.

Training and Development
The proposed Training and Development programme for a Doctoral Fellowship should meet the following criterion:

- The suitability of the proposed training to the needs of the project
- The extent to which the proposed practice activities will support the development of the applicant as a (practitioner) academic
- The extent to which the training and development plan will support the career aspirations of the applicant

Score (1-6)	5

Reason for your Score

Please provide the reason for your score and how it relates to the applicant's performance against the assessment criteria. This information will be sent to applicants as feedback.

Well rounded training programme

Significant issues that need clarifying at interview (if relevant)

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Selection Committee Assessment 2

Please read the application and provide a score for each assessment section using the following scale:

Person
Applicants for a Doctoral Fellowship should meet the following criteria:

- Relevant previous research experience and/or training suitable for undertaking a PhD in applied health or social care research
- Outputs from previous research experience and training relative to career stage and background
- Potential to become a future leader in applied health or social care research

Score (1-6)	4

Reason for your Score Please provide the reason for your score and how it relates to the applicant's performance against the assessment criteria. This information will be sent to applicants as feedback.
The individual had a long varied and relevant background. A slight concern with some experience cited was it was hard to determine whether it was in the last 5 years or much much longer ago Good publications. But seemingly no research grants

Research Proposal
Applicants for a Doctoral Fellowship should meet the following criteria:

- Scope of research suitable for a PhD
- Suitability and standard of methodology and research design to answer research question being proposed
- Likely impact of the research to patients/service users, carers, the public and health and/or care services
- Quality of doctoral research already undertaken (if applicable and/or where relevant)
- Appropriate and meaningful patient/service users, carer and public involvement
- Quality and scope of review of existing evidence

Score (1-6)	5

Reason for your Score Please provide the reason for your score and how it relates to the applicant's performance against the assessment criteria. This information will be sent to applicants as feedback.
Good PE Its not clear how outcome and resources / costs will be measured. This requires more detail discussion and time spent outlining methods here and where it fits into the research timeline. I'm unclear where the gold standard is coming from to measure all these improvements, for example the 10% better optimisation. I understand factors upset good delivery but at least spend some time defining ideal delivery. I agree with personalised medicine and targeting but I would still like to know at the start what optimal by CKD stage / level is. I think the stages and steps make sense, but the lack of baseline is confounding the stages. I think the author is intending to normalise and analyse existing data to give the baseline for deviation, But they need to explain this

more clearly.

In WP3 it becomes obvious of the importance that national guidance will play, but this needs saying and outlining earlier.

I'm very uneasy with work package 3 which is the model. I feel the author totally underestimates how difficult it will be to estimate a heterogeneous population and its difference from a systematic review and that the key patients variable and probabilities will happily drop out and into this model. Every step of the model (sensible steps) falls into neat 3 month chunks – this is ambitious. Cost and outcome measurement methods and approaches are not detailed.

The author clearly intends and has done a lot of work with PPI but I'm really struggling to work out where they actually fit at each stage and work package.

Site and Support
Applicants for a Doctoral Fellowship should meet the following criteria:

- Suitability and experience of supervisory team
- Standard and level of host organisation's support including evidence of supporting other early career researchers
- Track record of Host Department(s) in the research area being proposed
- Evidence of the host organisation(s) commitment to creating and maintaining an inclusive and supportive research culture, including evidence of commitment to the principles of equality, diversity and inclusion and research integrity

Score (1-6)	6

Reason for your Score

Please provide the reason for your score and how it relates to the applicant's performance against the assessment criteria. This information will be sent to applicants as feedback.

The supervisory team is an excellent fit and mix.

Training and Development
The proposed Training and Development programme for a Doctoral Fellowship should meet the following criterion:

- The suitability of the proposed training to the needs of the project
- The extent to which the proposed practice activities will support the development of the applicant as a (practitioner) academic
- The extent to which the training and development plan will support the career aspirations of the applicant

Score (1-6)	5

Reason for your Score

Please provide the reason for your score and how it relates to the applicant's performance against the assessment criteria. This information will be sent to applicants as feedback.

The training is very QM focused – well solely, apart from the overseas visit.

Data wrangling is an odd term and its not clear exactly what it means.

Its hard to determine exactly how long the courses will last.

Is there any other course in the UK worth doing?

I would like further justification of Canada over and above any other research institution.

Significant issues that need clarifying at interview (if relevant)
I'm unclear where the gold standard is coming from to measure all these improvements, for example the 10% better optimisation. Its not clear how outcome and resources / costs will be measured. This requires more detail / discussion and time spent outlining methods here and where it fits into the research timeline. I think the stages and steps make sense but the lack of baseline is confounding the stages.

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Summary

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Selection Committee Assessment 3

Please read the application and provide a score for each assessment section using the following scale:

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Applicants for a Doctoral Fellowship should meet the following criteria:

- Relevant previous research experience and/or training suitable for undertaking a PhD in applied health or social care research
- Outputs from previous research experience and training relative to career stage and background
- Potential to become a future leader in applied health or social care research

Score (1-6)	4

Reason for your Score Please provide the reason for your score and how it relates to the applicant's performance against the assessment criteria. This information will be sent to applicants as feedback.
Applicant is 20 years post primary medical qualification and has a mainline career as a GP in NE London with an interest in renal medicine - he is clinical lead for the ICB in this domain. He has been working as a research fellow out of QMUL for 2 years and has a post-graduate MSc in Public Health. 4 first author publications. Overall slightly behind curve relative to career stage, but recognise that he has been a clinician for much of the last 20 years. Has potential for future leadership in the domain of public health, especially as he already holds leadership roles in primary care, hence research experience and evidence from the proposed research can directly be used by applicant to foster service transformation.

Research Proposal
Applicants for a Doctoral Fellowship should meet the following criteria:

- Scope of research suitable for a PhD
- Suitability and standard of methodology and research design to answer research question being proposed
- Likely impact of the research to patients/service users, carers, the public and health and/or care services
- Quality of doctoral research already undertaken (if applicable and/or where relevant)
- Appropriate and meaningful patient/service users, carer and public involvement
- Quality and scope of review of existing evidence

Score (1-6)	4

Reason for your Score Please provide the reason for your score and how it relates to the applicant's performance against the assessment criteria. This information will be sent to applicants as feedback.
Research proposal includes: - systematic review construction reviewing evidence for various interventions' effectiveness in preventing CKD progression and CVD events - big data study looking at cohort characteristics of a population with CKD in NE London. - cost effectiveness study utilising Markov modelling to estimate ICERs arising from hypothetical strategies including 10% and 100% medication optimisation. Scope of research proposal is appropriate for a PhD and is appropriate for the 45 month length of the programme.

Methodology of the proposed research is standard and would anticipate no difficulties in completing.

Likely impact of research on patients in NE London, if completed successfully, would potentially be high, particularly as the applicant is in a clinical leadership role that would enable them to implement changes in NE London. It is not clear if the model findings might be applicable to areas outside - probably they would be in other areas of London, but the model would require new data inputs outside of London.

Although it is noted that PPIE support was used for development of the project, the exact manner in which the PPI's group perspectives contributed was not specified. The proposed PPIE strategy for project execution is reasonable.

Site and Support

Applicants for a Doctoral Fellowship should meet the following criteria:

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Score (1-6)	5

Reason for your Score

Please provide the reason for your score and how it relates to the applicant's performance against the assessment criteria. This information will be sent to applicants as feedback.

The supervisory group is fully experienced in the requisite techniques and in doctoral supervision, so anticipate few problems from this end. Applicant has also assembled a collaborative group who should be able to supply specific expertise to assist the applicant in execution of the project. The host department (Wolfson Institute within QMUL) has the requisite expertise and support in this area - the proposal is fairly mainstream for this institution. Bona fides with regards to commitment to EDI are given, but not specifically for research integrity.

Training and Development

The proposed Training and Development programme for a Doctoral Fellowship should meet the following criterion:

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Score (1-6)	4

Reason for your Score

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The proposed T&D programme covers a reasonable blend of project-specific training, generic research skills training, and also career development training appropriate to the career level of the applicant (based on the NIHR Leaders Support and Development Programme). It includes an overseas research visit which should bring new perspectives to the applicant's experience.

Significant issues that need clarifying at interview (if relevant)

- similar exercises in characterising cohort characteristics of people with CKD have already been done (e.g. 10.1136/bmjopen-2022-065927 which used CPRD data, and this is not in the reference list). How will the proposed research project add to information from this?

- applicant needs to justify using data only from NE London - would it not be more informative to consider information as well from other areas of London or even nationwide to increase applicability. Is NE London data generalisable to other areas of UK? Will the model generated be extensible?

- proposal is not specific as to which population with CKD will be studied/modelled? eGFR <60 (G3a+) or <45 (G3b+)? Albuminuria classes A1+ or A2+?

- research proposal refers to 'subgroups' but these are never defined? This is of key importance, as the various interventions proposed may have conditions attached to them in NICE guidance that means that subgroups are treated differently, e.g. NICE TA775 recommends dapagliflozin for people with eGFR 25-75 with T2D or with UACR of >22.6 mg/mmol (which unfortunately does not entirely align neatly with standard CKD classification groups). GLP-1RA NICE guidance generally recommends for people with BMI >35 only - in case of diabetes, this is after patients are not adequately controlled with 3+ agents. Any cost-effectiveness modelling has to take account of the landscape of approvals/guidance.

- is a Markov model appropriate, or would microsimulation be more appropriate?

NIHR | National Institute for Health Research

DF R11

Shortlisting Review Form

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Applicant	Dr Chris Carvalho
Applicant Organisation	Queen Mary University of London
Total Amount Requested	£936,127.00
Reviewer Number	

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1. Review

1. Review

Strengths

- He is looking at a difficult area from a primary care perspective for which he should be commended.
- He is well placed in London to access a large patient group to furnish his model.
- He recognises he's not there on modelling yet but sees the approach is to get that experience and with strict times lines to keep himself on track.
- The PPI was good, can see PPI involvement but it is a small group
- There are a lot of courses alongside everything else most seem ok but very Queen Mary centric apart from a long and expensive Canadian visit.
- The supervisory team is an excellent fit and a good mix.

Weaknesses
Please summarise the committee's criticisms of the application, recording any suggestions for improvement.

- He's doing a modelling course but realises this does not make him a modeller.
- There are considerable risks that he will not be able to get all the outcome and resource data needed for the model and it was not clear that there were real back up plans to construct it if the literature couldn't provide it.
- The overseas visit is long and expensive and not sure its totally justified.
- Training plan is concerning due to the number of courses and might be overstretching himself.

CONDITIONAL OFFERS ONLY
Please stipulate the condition(s) to be met.
Please only detail conditions that were agreed by the Committee at the interview meeting.

The 6 month visit needs to be better justified, whilst an overseas trip to gain knowledge is fine when its gaining key skills, but less justified if a large proportion of these key skills can be delivered domestically.

No previous online reviews in correct status have been found.